

1. PHASE 3, RECONCILED: WHAT THE GEOMETRY BUYS AND WHAT IT DOES NOT

Phase 3 is where the algebraic drop-in of Level 1 meets empirical descent, and it is the one place in this pipeline every claim can be put against a control. We report it in that spirit: three of the geometric constructs survive measurement in a precise, bounded form; one—the strip energy E —does not, and is retired here. The organising fact is a single measurable clock, the *block gradient-profile drift*

$$\text{drift}(t) = \|\pi(t) - \pi(t - w)\|, \quad \pi_a = \|\nabla_{\theta_a} \mathcal{L}\| / \sum_b \|\nabla_{\theta_b} \mathcal{L}\|,$$

the rate at which the interaction graph of §?? is still changing. Everything Phase 3 does—when to stop, when to jump, what to freeze—is timed off this one signal.

1.1. Crystallisation is real, as a regime: why the Snapper jump works. The crystal picture names a genuine phenomenon, provided it is stated as a property of the *settled* trajectory rather than of the whole descent. Once $\text{drift}(t) \rightarrow 0$, the parameter path becomes locally *low-order polynomial* in the step index: a degree- ≤ 2 least-squares fit over a short window has relative residual $\sim 10^{-3}$. This is the operational content of “crystallisation” — the descent has stopped searching and is now a deterministic, algebraically-predictable curve. It is exactly the condition Snapper’s polynomiality (§5) requires, and where the approach is still *steep*, the extrapolation lands: the 1-shot jump to the fitted minimum recovers the descent it skips (validation $0.066 \rightarrow 0.015$ across the alignment approach).

The bound, stated plainly because it is where the naive reading fails. On a *flat* basin floor the same fit is just as clean (10^{-3} residual) but the jump *overshoots*: extrapolating a settled-but-flat trajectory forward moves the weights along a direction that no longer lowers the loss (measured $+0.4$ validation on every attempted jump; 0% of steps culled). Crystallisation therefore licenses the jump only on a steep, clean-basin descent; on the plateau the correct crystallised operator is not the jump but the *stop*. The Snapper is a steep-regime move; E was never what governed it.

1.2. The geo-stop, measured: drift replaces the E -based criterion. The geometry does converge before the loss — but the operative signal is $\text{drift}(t)$, not a level set of E . On the real corpus, drift falls $0.097 \rightarrow 0.013$ while the loss is still creeping down; a stop triggered on a sustained low-drift window ends training at $\sim 50\%$ of the step budget for a few-percent validation cost. This is a strictly cheaper geo-stop than the $\Phi_{cl}/\tau/r_{m2}^\sigma$ stack — one gradient-norm profile per step, no SVD, no Hessian-strip correlation — and unlike E it survives its control (it moves, monotonically, and it moves before the loss does). The gluing defect $\tau = \|\nabla_{FF}\|/\|\nabla_{Emb}\|$ is kept: it is a real, monotone basin coordinate. The strip energy E is not kept.

1.3. What the basin is made of: the cull ledger. The settled basin has spatial structure, and it is the interaction graph of §??: an Emb/FF hub with attention peripheral early, densifying late on corpora that recruit it. This makes Phase 3’s optimiser cost *partly* reducible, and the reduction is bounded, not total.

- **Cullable periphery.** As drift settles, the value/output projections’ preconditioner flattens ($\text{CV}(1/\sqrt{v})$: $0.57 \rightarrow 0.24$), and W_V, W_O run on scalar SGD at $\sim$ free validation cost — 18% of parameters leave the second-moment state. The cullable fraction *rises* as drift falls: spatial and temporal cullability are the same clock.
- **Irreducible core.** Emb, W_Q , W_K , and FF keep dispersed, live preconditioners throughout; scalar-SGD on all of them fails ($+103\%$), freezing the whole core diverges ($\times 10^2$ loss). The adaptivity here is load-bearing, and no amount of settling removes it.
- **Unresolved (Q/K frozen shape).** Freezing W_Q, W_K ’s second moment—keeping the dispersed diagonal but culling its update—costs between $+0.7\%$ and $+8.2\%$ across two seeds, i.e. at the noise floor. Whether this lever exists is a variance measurement we have not yet closed; it is reported as open, not as a win.

1.4. The strip energy E , retired. $E = \sum_k \sum_i \arccos \sigma_i(U_k^\top U_{k+1})$ is a property of the *construction*, not of the trajectory. It is near-constant because it is built from the near-degenerate singular directions of W_K (measured gaps ~ 0.008), whose arccos is pinned by the Mean-Field pump; its automatic gradient is wrong by 6.4% for exactly this reason. At $P \approx 4.3 \times 10^6$ a random direction has $|\cos(\dot{\theta}, \nabla E)| \approx P^{-1/2} \approx 5 \times 10^{-4}$, and the trajectory’s value is the same order: the instrument has no power to distinguish “follows the level set” from “ambient.” We therefore make *no* dynamical claim for E . It was a coordinate of the mould, not of the motion. This retirement is the price of keeping the rest honest.

1.5. What the “manifold” claim can bear. The manifold is real in two senses and unreal in a third, and the distinction matters. It is real as the *constrained* geometry the descent lives on — the LayerNorm hypersphere S^{D-1} , the Softmax simplex — and real as the reproducible *low-rank tangent structure* of the trajectory: five optimisers’ local tangent spaces collapse to an effective rank 3.44/5 in two near-orthogonal families, a result three independent instruments agree on. It is *not* real as a small function-space quotient one could optimise on: the Jacobian $J_\theta = \partial f / \partial \theta$ is isotropic (no resolved low rank), so there is no visible subspace to collapse the descent onto. “The trajectory is a low-rank constrained object” holds; “solve on the quotient” does not.

TABLE 1. Phase 3 claims against their controls.

Claim	Status	Control / evidence
Geo-stop before loss plateau (drift $\rightarrow$ 0)	holds	drift 0.097 $\rightarrow$ 0.013 while loss creeps; real model
Snapper jump, steep clean-basin approach	holds	val 0.066 $\rightarrow$ 0.015; fit residual 10^{-3}
Snapper jump, flat plateau	fails	overshoots +0.4 val, 0% steps culled
$W_V, W_O \rightarrow$ scalar SGD (spatial cull)	holds (~free)	cull% 0 $\rightarrow$ 18 as drift falls
Core (Emb/ W_Q / W_K /FF) reducible to SGD	fails	SGD-all +103%; core-freeze diverges
Q/K frozen-shape (cull v -update)	unresolved	+0.7% vs +8.2%, two seeds
Star $\rightarrow$ dense interaction graph	holds (task-dep.)	r : 0.57 $\rightarrow$ 0.97 real corpus; $\rightarrow$ 0.04 Markov
2-family low-rank tangent structure	holds	eff. rank 3.44/5; three instruments
E as conserved dynamical invariant	fails (artifact)	ambient; $ \cos \approx P^{-1/2}$; arccos of degenerate σ
Low-dim function quotient (“solve on it”)	fails	Jacobian isotropic

1.6. Ledger. The reading that survives: Phase 3 crystallises a *predictable* trajectory on a *constrained, low-rank* basin, and that predictability is what the Snapper exploits and what the drift-stop detects. What does not survive is the claim that any single scalar (E) or small quotient carries the dynamics. The crystal is real; the mould-coordinate E was mistaken for it.